

pop quiz - torques - equilibrium

 This is a preview of the published version of the quiz

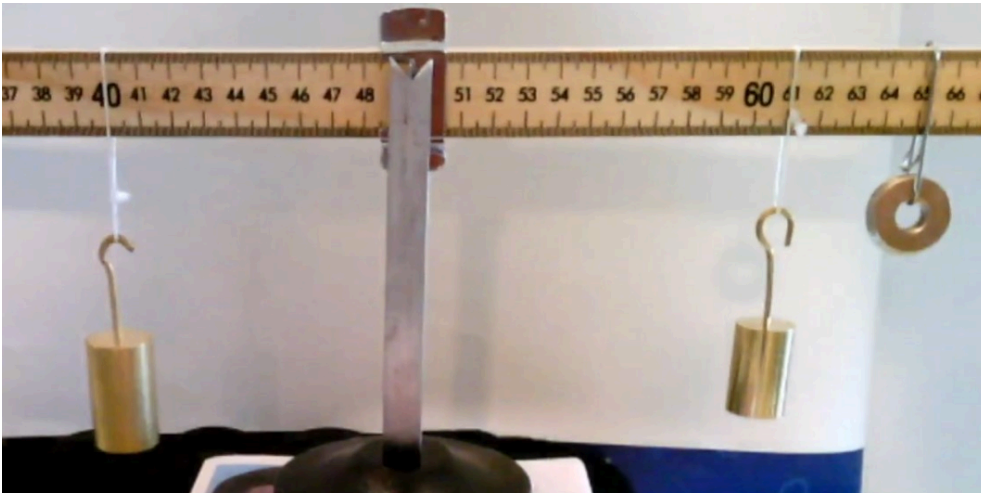
Started: Apr 29 at 8:01pm

Quiz Instructions

Talk to each other



Question 1 14.3 pts



Here is a static equilibrium. from left to right. you see that m_1 is 10cm from the fulcrum - left side
 m_2 is at 11cm from the fulcrum right side. and m_3 is at cm from the fulcrum right side. count the cm.

m_1 is 50g. m_2 is 25 g. so what is m_3 ?

Don't ask me. Use $\sum(CW) = \sum(CCW)$

Don't convert g to N. don't convert cm to m

☐ 15g

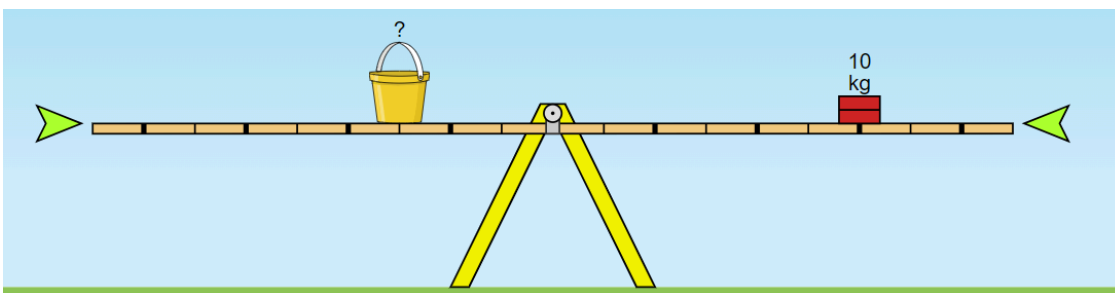
☐ 25g

☐ 50g

☐ 5g

☒

Question 2 14.3 pts



what is the mass of the bucket?

☐ 20kg

☐

10kg

☐

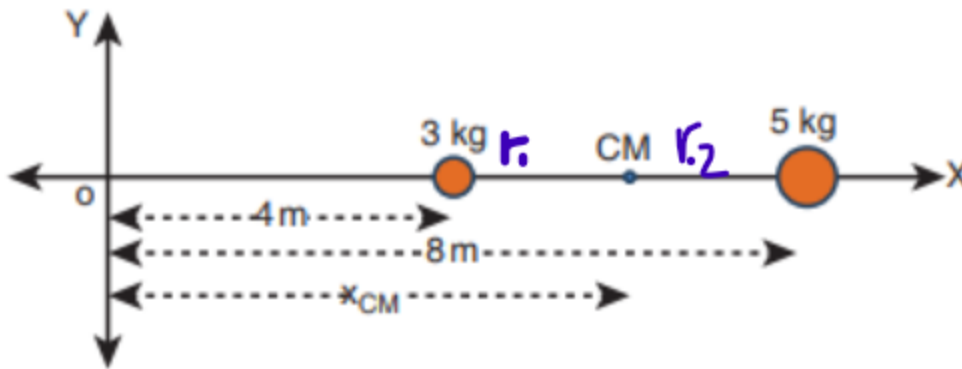
30kg

☐

5kg

☐

Question 3 14.3 pts



Find the position of the center of mass. $x_1 = 4\text{m}$ and $m_1 = 3\text{kg}$. $x_2 = 8\text{m}$ and $m_2 = 5\text{kg}$.

If you place a fulcrum at CM it will be in equilibrium ($3r_1 = 5r_2$). With a little bit of algebra you get the equation for the center of mass:

$$X_{cm} = (x_1 m_1 + x_2 m_2) / (m_1 + m_2)$$

☐

6.5cm

☐

6cm

☐

7cm

☐

5cm

☐

Question 4 14.3 pts

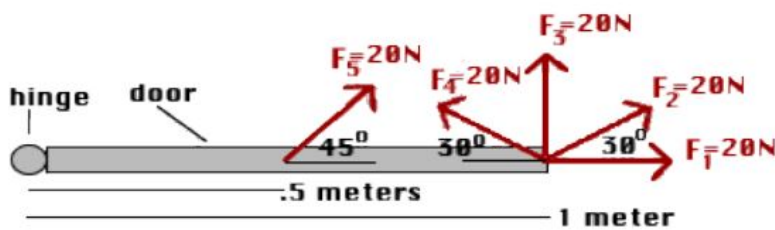


Its an equilibrium because

- ☐ his center of mass is above the line connecting the hands
- ☐ he is held up by an insisible string. can't be otherwise
- ☐ he is extremely muscular
- ☐ this is a photo generated by Grok (X AI). can't be



Question 5 14.3 pts

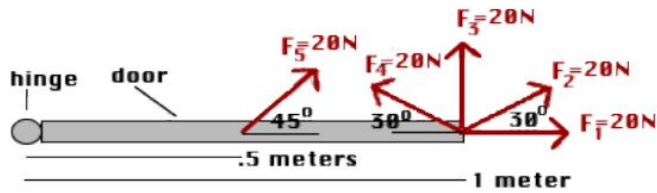


The torque of the force F1 is

- ☐ 0
- ☐ 20N.m
- ☐ 10 N.m
- ☐ I miss the class



Question 6 14.3 pts

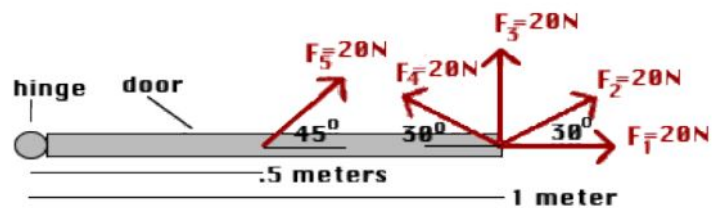


The torque of the force F_3 is

- ☐ 20N.m
- ☐ 0
- ☐ 10 N.m
- ☐ 5N.m



Question 7 14.3 pts



The torque of F_5 is

hint: Use the perpendicular component (ry-component of F_5).

- ☐ 7N.m
- ☐ 14 N.m
- ☐ 20N.m
- ☐ 0

Not saved

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